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Supplementary Validation: Resolution-dependent correspondence in DNA-damage models

Carlos Ramirez Ovalle

Pontificia Universidad Javeriana Cali, Colombia

This supplement retains implementation-verification and exploratory-data details supporting the main biological comparison. It is not independent biological validation of the curated GIM models.

S1. Formal and software verification

S1.1. *Construction-level software benchmark and comparator study*

Six deterministic pairs were specified before any biological application. They provide behavioral-class ground truth by construction and test whether the software recovers intended formal distinctions; they do not provide biological ground truth. A 12-step labeled workflow supplies identity, controlled silent subdivision, label mismatch, label-order swap, and an added observable branch. The sixth pair is $a.(b+c)$ versus $a.b + a.c$ from Example 1 of the main article. Table S1 gives construction-level and computed classes.

On this construction-level decision task, weak bisimulation recovered all six binary weak-equivalence decisions. Truncated trace equality recovered 5/6 but accepted the trace-equivalent branching pair; LTS-GDA recovered 4/6, the structural profile 3/6, and direct PN-GDDA 2/6. PN-GDDA assigned 1.0 to identity, silent refinement, label mismatch, and order swap, while added-branch and branching pairs scored 0.9676 and 0.9657. No threshold over the six PN-GDDA scores exceeded 4/6 because two negative controls tied the positive controls at 1.0. Figure S1 reports all predeclared decisions.

The structural profile averages state-count, edge-count, label-multiset, and out-degree-multiset similarities. The trace diagnostic declares a depth-8 match when $d_8 = 0$. PN-GDDA was implemented from its published definition using 151 directed bipartite graphlets and the 592 published/Holmes orbit slots.¹ For orbit j ,

$$N_G^j(k) = \frac{D_G^j(k)/k}{\sum_{r>0} D_G^j(r)/r}, \quad A^j(G, H) = 1 - \frac{\left[\sum_k (N_G^j(k) - N_H^j(k))^2 \right]^{1/2}}{\sqrt{2}}, \quad (\text{S1.1})$$

Table S1. Controlled benchmark; the expected class is not used by the classifier.

Case	Change	Expected	Observed
Identity	exact copy	strong equivalence	strong equivalence
Silent refinement	insert internal steps	weak equivalence	weak equivalence
Label mismatch	replace an observable	not comparable	not comparable
Order swap	preserve counts, change order	not comparable	not comparable
Added branch	add an outcome	reference in candidate	reference in candidate
Early/late choice	move decision point	candidate in reference	candidate in reference

and PN-GDDA is the mean A^j . Synthetic LTSs use an exact state-machine Petri net encoding; curated pairs use their native nets. The 0.9 threshold is a fixed diagnostic decision, not an equivalence relation. LTS-GDA instead combines four reachable-graph orbits with labeled edge, two-step, divergence, and convergence motifs.

The catalog audit recovered 151 graphlets and 592 slots from hash-verified Holmes 1.1.1/2.0.1.2 JARs. A direction- and type-preserving automorphism partition has 576 distinct orbits; using it is reported only as sensitivity. A Java probe matched every inspected Holmes 1.1.1 topology/root assignment. On a four-node test net and its one-arc deletion, Python and Holmes scores were 0.9872027465870733 and 0.987202746587073, agreeing to 12 decimal places. Holmes is an independent software reference for the structural score.²

S1.2. *Independent formal oracles*

Each analyzed LTS was exported in Aldebaran AUT format and evaluated with `ltscompare` from hash-pinned mCRL2 202607.0.³ The tool independently checked strong bisimilarity, weak bisimilarity, and exact weak-trace equivalence with `tau` hidden. Its interface does not expose a general weak-simulation preorder. For the tau-free Example 1 and middle strictness witness only, R2 additionally uses its strong-simulation preorder, which coincides with weak simulation when neither input has silent transitions; both directions agree. No such weak-preorder claim is made for systems containing silent transitions.

A separately coded attacker–defender game grows the least set of positions from which an unmatched move can be forced. It agreed with the production simulation algorithm on all 67,600 ordered pairs among all 260 one- and two-state LTSs over $\{a, \tau\}$. This is exhaustive within that finite universe, not a proof for arbitrary systems. Across the six synthetic pairs, six public-model controls, and nine HPN comparisons below, Python and mCRL2 agreed on all 42 reported strong/weak decisions. Theorem 1 of the main article, finite exhaustive testing, and external-tool comparison are distinct evidence layers.

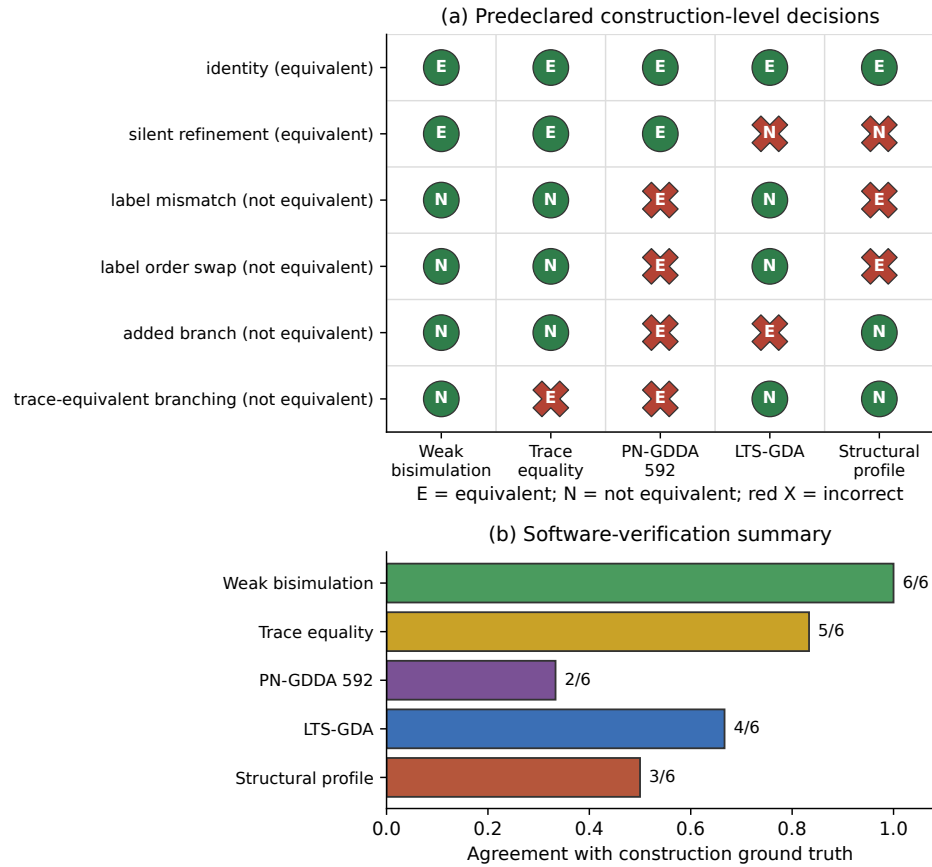
Controlled benchmark: software verification only

Fig. S1. Controlled software/formal benchmark. (a) Decisions for six cases whose labels were fixed by construction; a red cross marks disagreement with that label. (b) Agreement count on this limited task. Trace equality loses branching; PN-GDDA is label-blind; LTS-GDA retains local labels but does not abstract silent refinement or recursively match successors. The panel verifies implementation behavior and does not measure biological validity or generic method superiority.

S1.3. Observed computational scaling and reproducibility

Linear workflows of 8, 16, 32, 64, 96, and 128 transitions were compared with an identity and a silently refined copy. Median runtime over three repetitions rose from about 0.4 ms for nine states to about 0.6 s for the largest pairs. The 128-step identity relation had 16,641 candidate pairs; refinement increased it to 18,705 pairs over 129 and 145 states. Every class remained as constructed (the complete plot and table are archived in the repository). Timings used Python 3.11.8 on an Apple M3 Pro

Table S2. Public GINsim models and control outcomes. S/W/T denote strong, weak, and exact weak-trace equivalence.

Model	Format	States	Edges	Copy	Silent	Perturbed
Mammalian cell cycle	SBML-q	826	3,413	Y/Y/Y	N/Y/Y	N/N/N
p53-Mdm2 damage	GINML	17	26	Y/Y/Y	N/Y/Y	N/N/N

with 18 GiB memory and are machine-dependent.

The Python standard library implements reachability, formal relations, parsers, and controlled models. NumPy, pandas, Matplotlib, NetworkX, pydot, and Graphviz support tabulation and figures. Fixed seeds, source hashes, machine-readable outputs, unit tests, an executed notebook, environment files, and continuous integration are public. Runtime CSV values are excluded from byte-level regression because they depend on hardware; all scientific outcomes and other result tables are deterministic.

S2. Cross-format ingestion and execution controls

Two externally authored GINsim models test the complete import-to-comparison path: the 13-component mammalian cell cycle in SBML-qual and the four-component multilevel p53-Mdm2 model in GINML.⁴⁻⁷ Files, URLs, and SHA-256 values are archived. Unitary-asynchronous updates move one enabled component by one level and label its direction. Each base model was compared with an exact copy, five silent subdivisions, and a reachable-label perturbation. All six controls recovered their predeclared strong/weak/exact-trace outcomes in Python and mCRL2 (Table S2).

Synchronous execution reached 13 of 826 asynchronous cell-cycle states (Jaccard 0.0157) and eight of 17 p53-Mdm2 states (Jaccard 0.471), confirming sensitivity to update semantics. These are cross-format execution controls, not biological validation.

S3. Exploratory HPN-DREAM/CASPOTS held-out analysis

The public HPN-DREAM phosphoproteomic series and CASPOTS Boolean families contain 72 BT20, 191 BT549, and 21 MCF7 networks.^{8,9} One representative per cell line was selected before test responses were read: the model with maximum mean within-family Jaccard similarity between active clauses, with exact lexicographic tie-breaking. This structure-only rule selected zero-based rows 24, 137, and 9 (Table S3); an anti-leakage test forbids access to test files during selection.

The interface is the intersection of seven measured PI3K/MAPK/mTOR readouts. The three common test perturbations are mTOR inhibition, IGF1 plus mTOR inhibition, and 4EBP1 plus mTOR inhibition. Measured time-zero values initialize readouts; stimuli and inhibitors are clamped; other relevant nodes start at zero. Primary asynchronous LTSs label interface updates and hide all others. The nine

Table S3. HPN-DREAM families and structure-only representatives.

Cell line	Family	Medoid row ^a	Clauses	Mean Jaccard
BT20	72	24	29	0.815
BT549	191	137	25	0.854
MCF7	21	9	23	0.851

Note: ^aZero-based row; selected from model structure only.

cross-cell pair-condition comparisons were repeated synchronously; this stress test is not a most-permissive analysis. CASPOTS itself evaluates asynchronous trajectories.^{9,10}

Experimental distance is RMSE over common post-zero times and readouts. Spearman associations use an exact one-sided permutation of the three cell-pair labels within each condition ($3!^3 = 216$). CASPOTS/Clingo compatibility scores consume optimization to completion, use deterministic secondary tie-breaks, and were reproduced twice. Native specificity uses the exact sign permutation of three native-minus-mean-cross differences.

Selected LTSs had 2–44 states. None of the nine cross-cell pairs was weakly bisimilar: six had one-way simulation and three were not comparable. Direction was retained explicitly, with three left-model-in-right-model and three right-model-in-left-model relations. These categories do not form a justified one-dimensional order, and the held-out RMSE is symmetric between each cell pair. We therefore removed the former ordinal class coding and performed no scalar formal-class test. Descriptively, median RMSE was 0.2418 for the six one-way relations and 0.2071 for the three non-comparable pairs; after preserving direction, the corresponding medians were 0.2757, 0.1919, and 0.2071, each based on only three observations. These summaries are not inferential evidence.

The two predeclared continuous diagnostics were retained. Truncated-trace distance gave $\rho = -0.596$ (exact one-sided $p = 0.556$), and LTS-GDA distance gave $\rho = 0.669$ ($p = 0.097$) under 216 condition-stratified permutations. Neither association is conclusive. Synchronous updating preserved seven of nine formal classes.

Native medoid RMSE was 0.33024 (BT20), 0.31335 (BT549), and 0.27726 (MCF7), against discretization RMSE 0.32944, 0.30079, and 0.27726. Native excess-RMSE advantage over mean cross-cell medoids was 0.00212 and not established by the exact sign test ($p = 0.25$; Fig. S2). The result demonstrates an exploratory held-out compatibility check and separates formal similarity from empirical response similarity; it does not demonstrate cell-line specificity or predictive discrimination.

S4. Second-round quantifier and direction audit

The production implementation was not changed to conform to a proposed reinterpretation of simulation. Instead, the R2 audit independently constructs silent

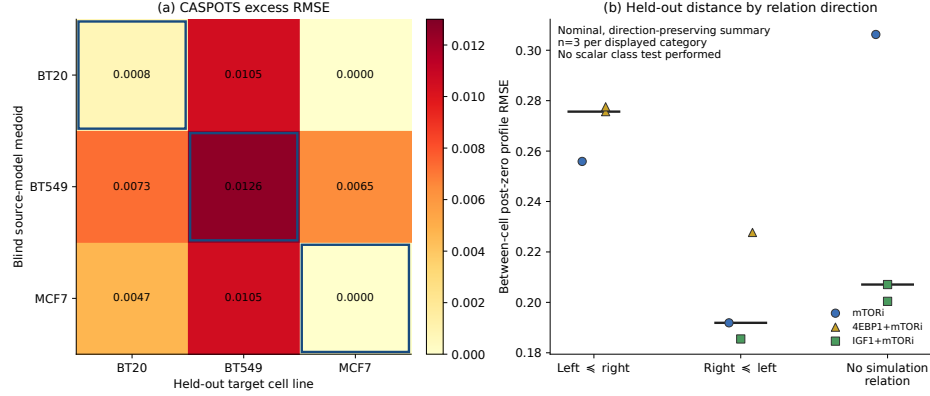
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Fig. S2. Exploratory held-out HPN-DREAM analysis. (a) CASPOTS excess RMSE by source medoid and target; outlined cells are native contexts. (b) Between-cell profile RMSE with the direction of weak simulation retained. Horizontal bars are medians; each category has $n = 3$, and no scalar formal-class test was performed. Neither panel establishes native cell-line specificity.

closures and weak targets from raw edges, and grows a least losing set in simultaneous attacker-defender rounds. It does not call the production adjacency, closure, weak-step, or relation routines. Exact weak-trace inclusion is checked by a finite epsilon-NFA subset-product traversal, treating all original states as accepting, without a depth cutoff.

For Example 1 in the main article, complete acyclic path enumeration also gives $\{\epsilon, a, ab, ac\}$ on both sides. The supplied five-pair early-to-late relation satisfies every transfer clause. The reverse calculation first removes (ℓ_1, e_b) because c cannot be matched and (ℓ_1, e_c) because b cannot be matched; the initial pair subsequently has no surviving reply. The machine-readable deletion certificate records these dependencies.

All three strictness witnesses and 32 ordered model pairs (six synthetic, three GIM interfaces, five curated modules, and 18 asynchronous/synchronous HPN comparisons) agree on strong/weak bisimilarity and both weak-simulation directions between the production code and independent audit. Every synthetic class still agrees with its construction-level expectation. Direction fields explicitly state `left_simulated_by_right` and `right_simulated_by_left`.

The complete one-/two-state universe contains 260 named LTSs over $\{a, \tau\}$ with initial state zero, hence 67,600 ordered pairs. The expanded audit found no simulation or weak-bisimulation disagreement and no failure of the hierarchy or trace-inclusion implications. Simulation is reflexive and transitive on this universe. There are 41,080 weakly bisimilar ordered pairs and 4,848 mutually simulated but non-bisimilar ordered pairs. No one-way simulation with exact trace equality occurs in this small universe; the larger acyclic Example 1 supplies that witness. The finite enumeration supports software correctness only; it does not replace the general

proofs.

Reproduce the new audit using `python -m src.formal_revision_audit` and `python -m unittest tests.test_formal_revision.R2 -v`. The four JSON/CSV reports in `results/` include the Example 1 certificate, hierarchy witnesses, all model-pair directions, and exhaustive counts. The main article’s appendix retains all definitions needed for its claims, the relation-deletion proof, and explicit proofs of the three failed converses. Detailed Petri nets remain in the main article rather than being removed during compression of validation material.

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